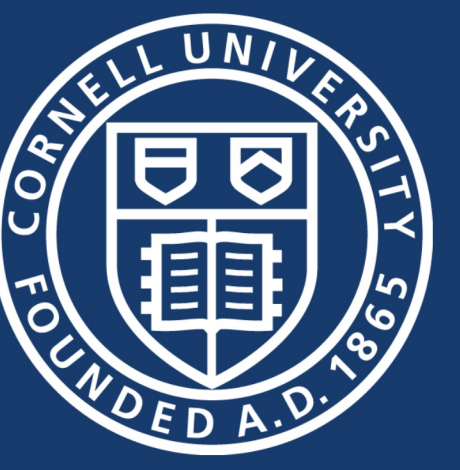


Unbounded Wind Fields for Stratospheric Balloon Simulation



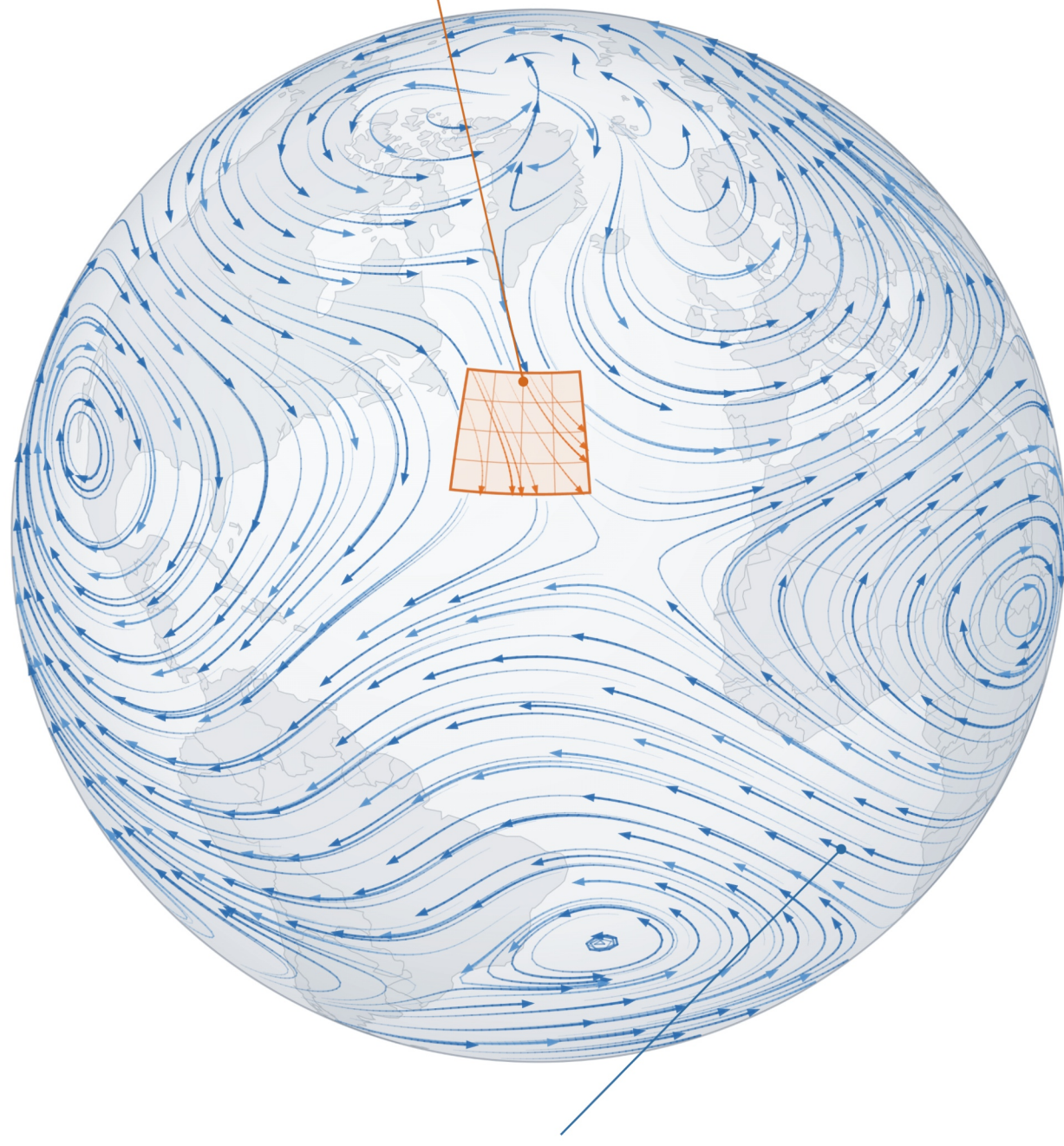
Shaurya Sen • Rohan Shankar

Faculty: Prof. Sarah Dean | Advisors: Vinamr Jain and Yujin Kim

1 Introduction

Balloon simulators need winds that are both **realistic** and **effectively unbounded**.

Previous work



Our method

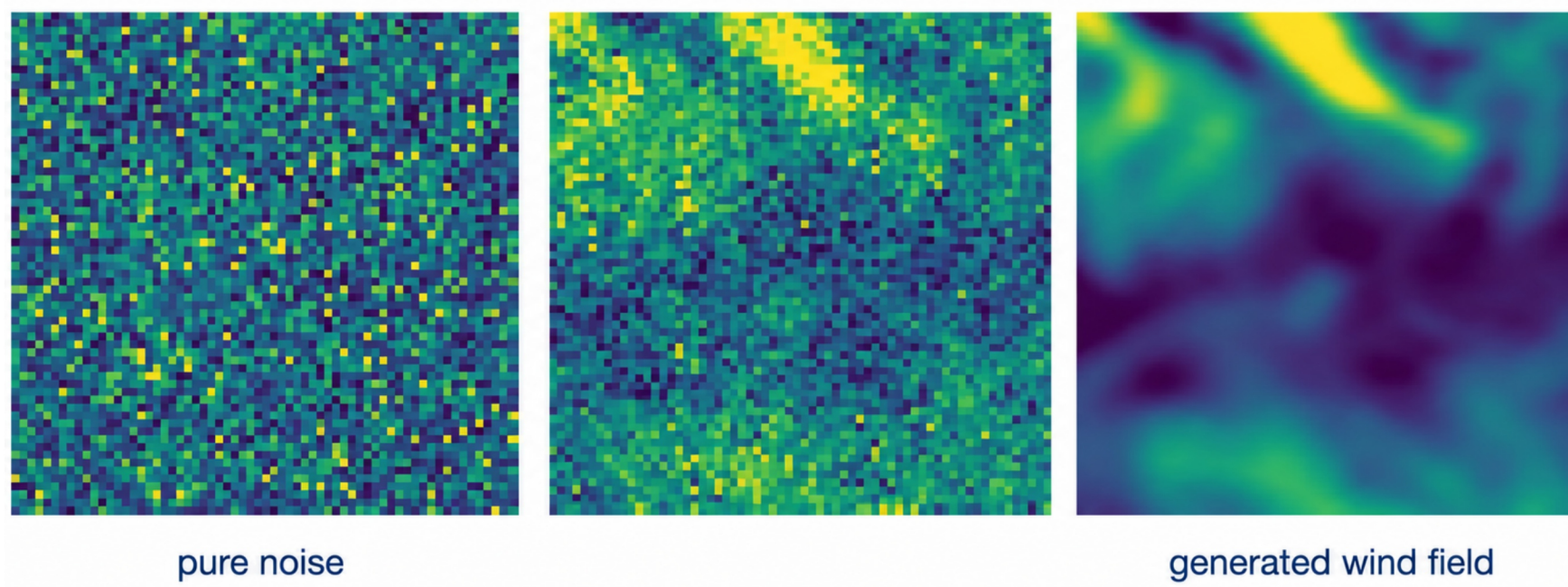
Lazily generate wind anywhere on the globe, across time

Our contributions

- InfiniteDiffusion for coherent, unbounded wind generation that outperforms prior balloon generators.
- A diffusion wind model conditioned on location and time.
- A realism evaluation spanning distributions, spatial physics, vertical structure, and time.

2 Base Diffusion Model

denoising



noising

Noisy wind volume

$64 \times 64 \times 18$ levels $\times (u, v) \times 4$ hours

Diffusion model

Conditioned on

Location: latitude / longitude grids
Time: annual + semiannual + hour-of-day
Noise: noise level σ

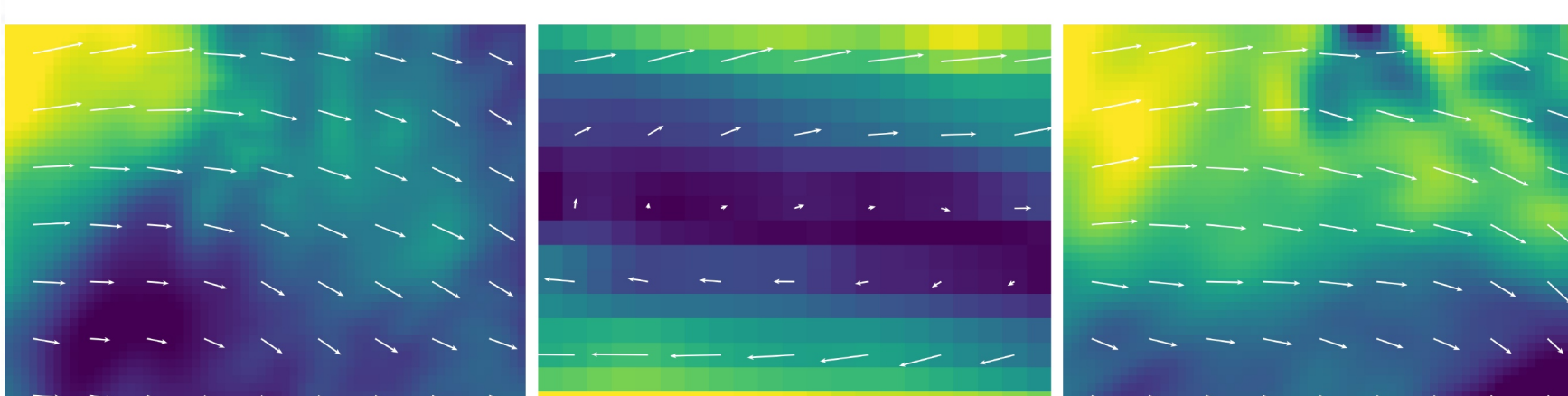
Generated wind volume

$64 \times 64 \times 18$ levels $\times (u, v) \times 4$ hours

ERA5

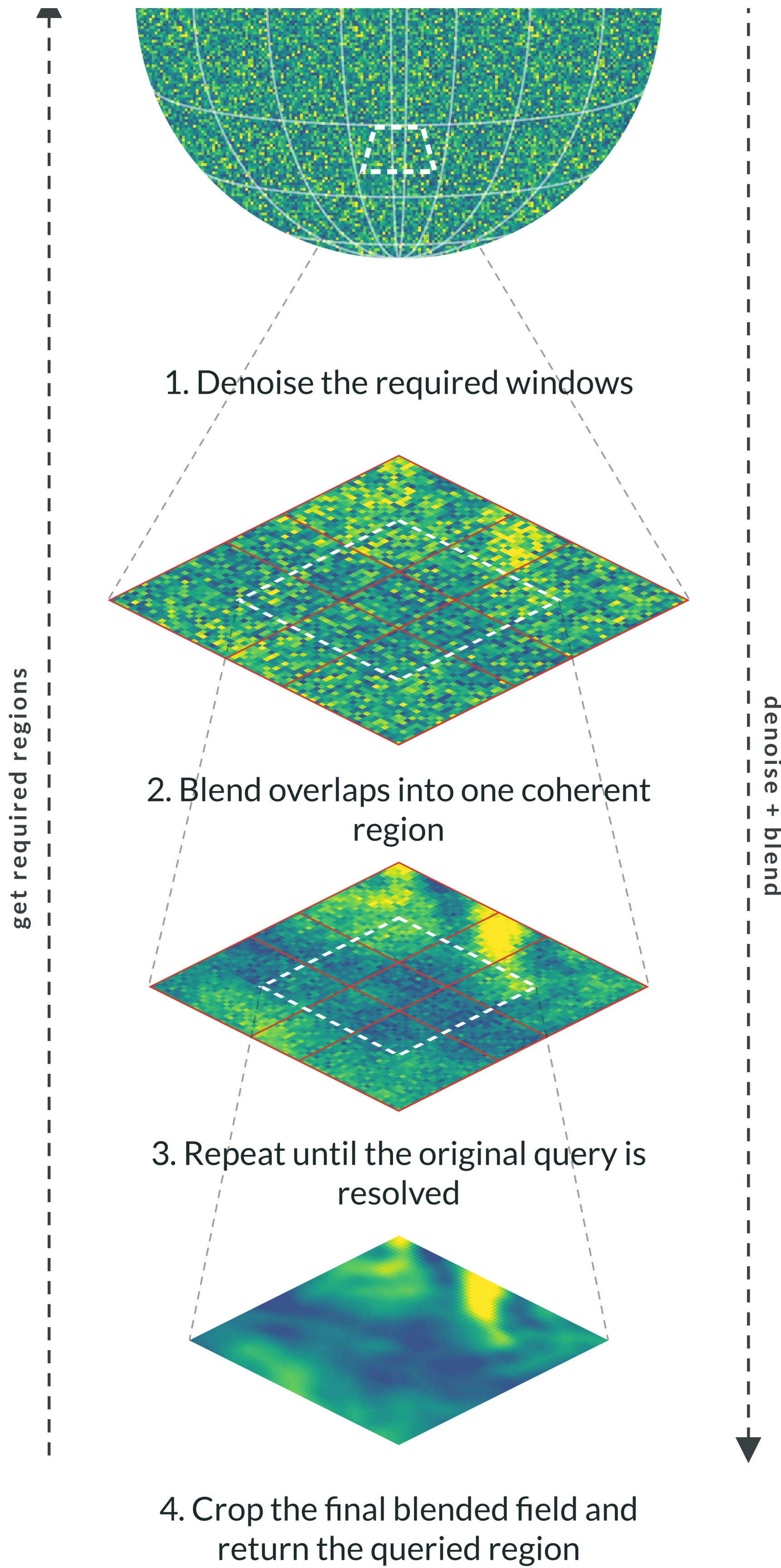
BLE-VAE

Conditional diffusion

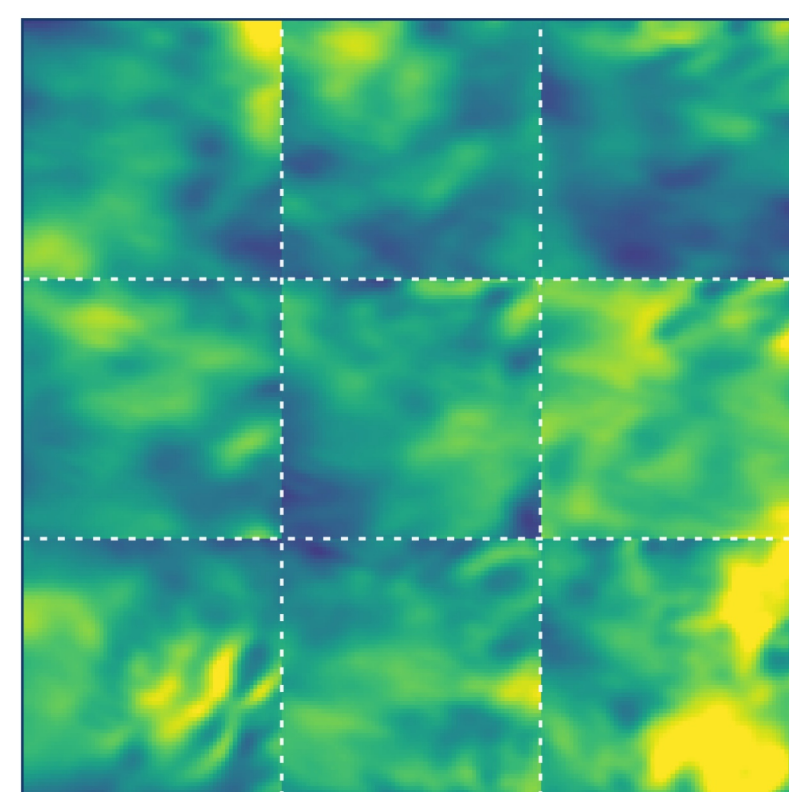


3 Infinite Diffusion

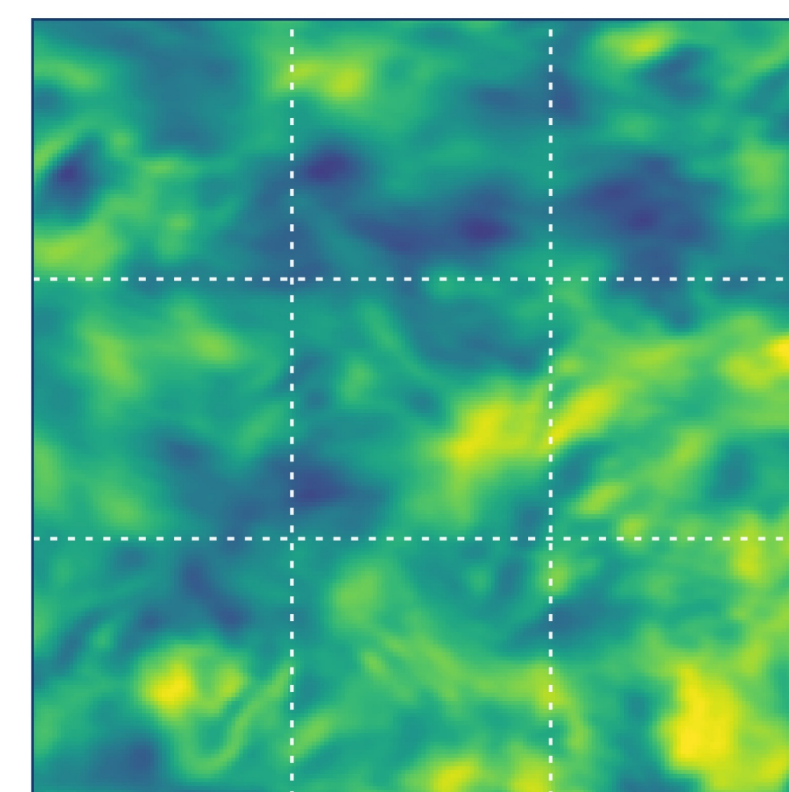
InfiniteDiffusion finds the windows a query needs, then denoises and blends them into one seamless field.



Independent tiling



InfiniteDiffusion

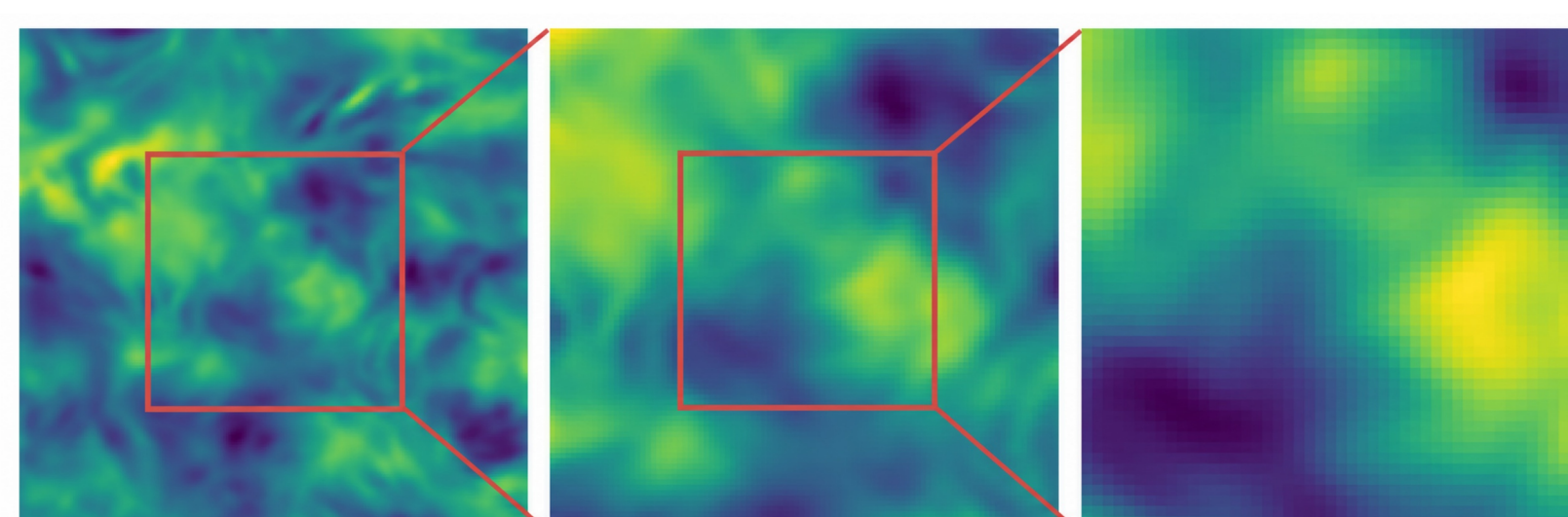


$O(1)$ random access

deterministic

unbounded

Deterministic random access in practice



4 Evaluation and Results

1. Distributions

Typical, conditional, and extreme winds.

2. Spatial physics

Energy by scale, divergence, and vorticity.

3. Vertical structure

Vertical shear across 18 pressure levels.

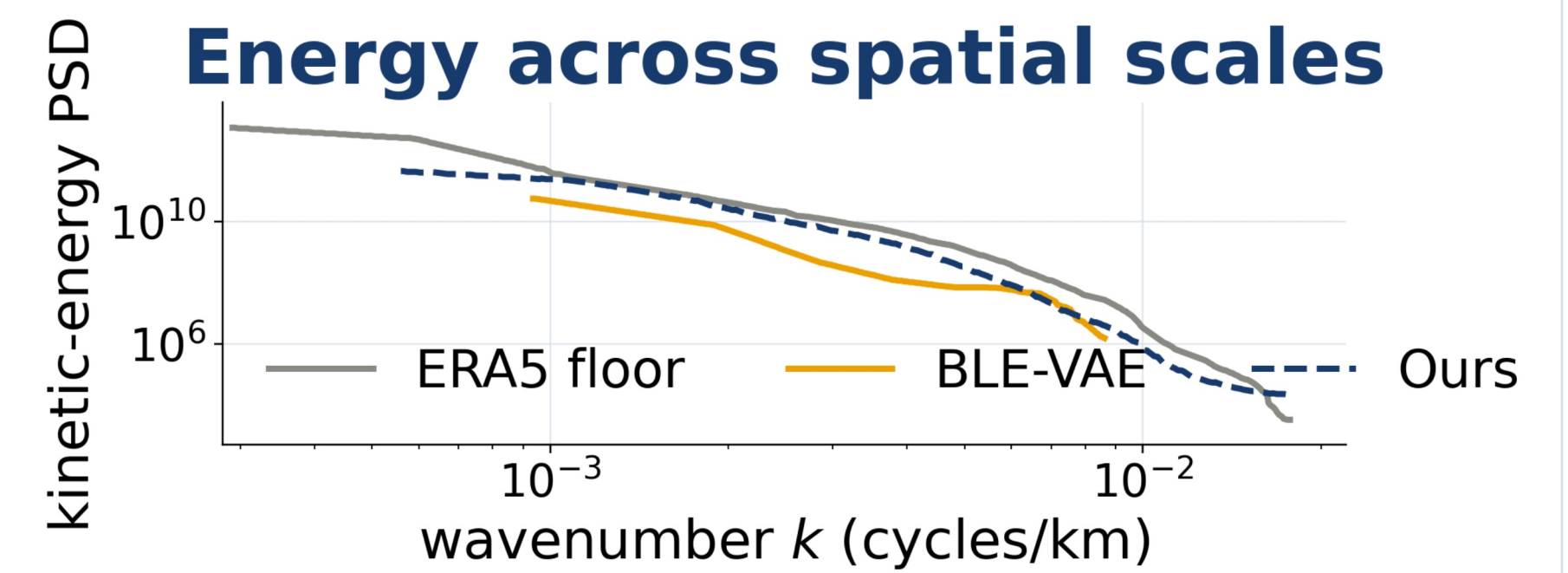
4. Temporal physics

Trajectory persistence and spread.

ERA5 floor: variation between held-out ERA5 subsets.

Results

TYPE	Metric	Floor	Simplex	BLE-VAE	Ours
DISTRIBUTIONS	$W_1(u c)$ (m/s) ↓	2.08	4.69	12.71	3.00
	$W_1(v c)$ (m/s) ↓	2.67	2.08	5.02	4.04
	Tail error 0.1% ↓	2.92	2.34	18.94	13.54
SPATIAL PHYSICS	SR_E ↓	0.25	0.75	2.73	1.53
	SR_{div} ↓	0.25	0.87	5.23	1.56
	SR_{vort} ↓	0.25	0.72	2.16	1.53
	L_{eff} (km) ↓	56 [†]	56 [†]	842	337
VERTICAL	W_1 shear u (m/s)/km ↓	0.20	0.31	3.62	1.00
	W_1 shear v (m/s)/km ↓	0.10	0.97	0.61	1.01
TEMPORAL	SR_{time} ↓	0.28	1.19	4.19	5.46
	Final spread ratio $\rightarrow 1$	1.04	0.53	0.11	0.20



What does SR_E measure?

Mean log-scale gap between model and ERA5 energy spectra. Lower is better.

What does Wasserstein distance measure?

Distance between ERA5 and model wind distributions at fixed location, month, and hour t . Lower is better.

What does L_{eff} measure?

Smallest wavelength where the model retains at least half of ERA5 energy. Lower is sharper.

What does W_1 shear measure?

W_1 between ERA5 and model distributions of $\Delta u / \Delta z$ and $\Delta v / \Delta z$. Lower is better.

What does trajectory spread measure?

Advect particles; compare their displacement and final spread with ERA5. Closer is better.

5 Conclusion

InfiniteDiffusion creates unbounded, conditional wind environments and outperforms prior bounded, unconditional balloon wind generators in our evaluation.

What worked

- Fast, infinite, lazy generation.
- Consistency across tiles.
- Space/time conditioning.

What remains

- Coarse conditioning for the base diffusion model.
- Examine how tile count affects realism and runtime.